

1. User Upcoming Events

```
SELECT u.full_name, e.title, e.city, e.start_date
FROM Users u
JOIN Registrations r ON u.user_id = r.user_id
JOIN Events e ON r.event_id = e.event_id
WHERE e.status = 'upcoming'
AND u.city = e.city
ORDER BY e.start_date;
```

2. Top Rated Events

```
SELECT e.event_id, e.title, AVG(f.rating) AS avg_rating
FROM Events e
JOIN Feedback f ON e.event_id = f.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(f.feedback_id) >= 10
ORDER BY avg_rating DESC;
```

3. Inactive Users

```
SELECT *
FROM Users u
WHERE u.user_id NOT IN (
    SELECT DISTINCT user_id
    FROM Registrations
    WHERE registration_date >= CURDATE() - INTERVAL 90 DAY
);
```

4. Peak Session Hours

```
SELECT e.event_id,
       e.title,
       COUNT(s.session_id) AS session_count
FROM Events e
JOIN Sessions s
ON e.event_id = s.event_id
WHERE TIME(s.start_time) >= '10:00:00'
AND TIME(s.end_time) <= '12:00:00'
GROUP BY e.event_id, e.title;
```

5. Most Active Cities

```
SELECT u.city,
       COUNT(DISTINCT r.user_id) AS registrations
FROM Users u
JOIN Registrations r ON u.user_id = r.user_id
GROUP BY u.city
ORDER BY registrations DESC
LIMIT 5;
```

6. Event Resource Summary

```
SELECT e.title,
       COUNT(CASE WHEN r.resource_type='pdf' THEN 1 END) AS pdfs,
```

```

        COUNT(CASE WHEN r.resource_type='image' THEN 1 END) AS images,
        COUNT(CASE WHEN r.resource_type='link' THEN 1 END) AS links
FROM Events e
LEFT JOIN Resources r ON e.event_id = r.event_id
GROUP BY e.event_id, e.title;

```

7. Low Feedback Alerts

```

SELECT u.full_name,
       e.title,
       f.rating,
       f.comments
FROM Feedback f
JOIN Users u ON f.user_id = u.user_id
JOIN Events e ON f.event_id = e.event_id
WHERE f.rating < 3;

```

8. Sessions per Upcoming Event

```

SELECT e.title,
       COUNT(s.session_id) AS total_sessions
FROM Events e
LEFT JOIN Sessions s ON e.event_id = s.event_id
WHERE e.status = 'upcoming'
GROUP BY e.event_id, e.title;

```

9. Organizer Event Summary

```

SELECT u.user_id,
       u.full_name,
       COUNT(e.event_id) AS total_events,
       SUM(CASE WHEN e.status = 'upcoming' THEN 1 ELSE 0 END) AS upcoming_events,
       SUM(CASE WHEN e.status = 'completed' THEN 1 ELSE 0 END) AS completed_events,
       SUM(CASE WHEN e.status = 'cancelled' THEN 1 ELSE 0 END) AS cancelled_events
FROM Users u
LEFT JOIN Events e
ON u.user_id = e.organizer_id
GROUP BY u.user_id, u.full_name;

```

10. Feedback Gap

```

SELECT e.title
FROM Events e
JOIN Registrations r ON e.event_id = r.event_id
LEFT JOIN Feedback f ON e.event_id = f.event_id
WHERE f.feedback_id IS NULL
GROUP BY e.event_id, e.title;

```

11. Daily New User Count

```

SELECT registration_date,
       COUNT(*) AS new_users
FROM Users

```

```
WHERE registration_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY registration_date;
```

12. Event with Maximum Sessions

```
SELECT e.title,
       COUNT(s.session_id) AS total_sessions
FROM Events e
JOIN Sessions s ON e.event_id = s.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(s.session_id) = (
    SELECT MAX(session_count)
    FROM (
        SELECT COUNT(*) AS session_count
        FROM Sessions
        GROUP BY event_id
    ) x
);
```

13. Average Rating per City

```
SELECT e.city,
       AVG(f.rating) AS avg_rating
FROM Events e
JOIN Feedback f ON e.event_id = f.event_id
GROUP BY e.city;
```

14. Most Registered Events

```
SELECT e.title,
       COUNT(r.registration_id) AS total_registrations
FROM Events e
JOIN Registrations r ON e.event_id = r.event_id
GROUP BY e.event_id, e.title
ORDER BY total_registrations DESC
LIMIT 3;
```

15. Event Session Time Conflict

```
SELECT s1.event_id,
       s1.title AS session1,
       s2.title AS session2
FROM Sessions s1
JOIN Sessions s2
ON s1.event_id = s2.event_id
AND s1.session_id < s2.session_id
AND s1.start_time < s2.end_time
AND s1.end_time > s2.start_time;
```

16. Unregistered Active Users

```
SELECT *
FROM Users u
WHERE u.registration_date >= CURDATE() - INTERVAL 30 DAY
AND u.user_id NOT IN (
    SELECT user_id
    FROM Registrations
);
```

17. Multi-Session Speakers

```
SELECT speaker_name,  
       COUNT(*) AS total_sessions  
FROM Sessions  
GROUP BY speaker_name  
HAVING COUNT(*) > 1;
```

18. Resource Availability Check

```
SELECT e.title  
FROM Events e  
LEFT JOIN Resources r  
ON e.event_id = r.event_id  
WHERE r.resource_id IS NULL;
```

19. Completed Events with Feedback Summary

```
SELECT e.title,  
       COUNT(DISTINCT r.registration_id) AS registrations,  
       AVG(f.rating) AS avg_rating  
FROM Events e  
LEFT JOIN Registrations r ON e.event_id = r.event_id  
LEFT JOIN Feedback f ON e.event_id = f.event_id  
WHERE e.status = 'completed'  
GROUP BY e.event_id, e.title;
```

20. User Engagement Index

```
SELECT u.full_name,  
       COUNT(DISTINCT r.event_id) AS events_attended,  
       COUNT(DISTINCT f.feedback_id) AS feedbacks_submitted  
FROM Users u  
LEFT JOIN Registrations r ON u.user_id = r.user_id  
LEFT JOIN Feedback f ON u.user_id = f.user_id  
GROUP BY u.user_id, u.full_name;
```

21. Top Feedback Providers

```
SELECT u.full_name,  
       COUNT(f.feedback_id) AS total_feedbacks  
FROM Users u  
JOIN Feedback f ON u.user_id = f.user_id  
GROUP BY u.user_id, u.full_name  
ORDER BY total_feedbacks DESC  
LIMIT 5;
```

22. Duplicate Registrations Check

```
SELECT user_id,  
       event_id,  
       COUNT(*) AS duplicates  
FROM Registrations  
GROUP BY user_id, event_id  
HAVING COUNT(*) > 1;
```

23. Registration Trends

```
SELECT DATE_FORMAT(registration_date, '%Y-%m') AS month,  
       COUNT(*) AS registrations  
FROM Registrations  
WHERE registration_date >= CURDATE() - INTERVAL 12 MONTH
```

```
GROUP BY DATE_FORMAT(registration_date, '%Y-%m')
ORDER BY month;
```

24. Average Session Duration per Event

```
SELECT e.title,
       AVG(
         TIMESTAMPDIFF(
           MINUTE,
           s.start_time,
           s.end_time
         )
       ) AS avg_duration_minutes
FROM Events e
JOIN Sessions s ON e.event_id = s.event_id
GROUP BY e.event_id, e.title;
```

25. Events Without Sessions

```
SELECT e.title
FROM Events e
LEFT JOIN Sessions s
ON e.event_id = s.event_id
WHERE s.session_id IS NULL;
```